Uniform Distribution

1 Definition

A random variable X is said to have Uniform distribution over the set $\{x_1, x_2, \dots x_N\}$ if $P(X = x_i) = c, \forall i = 1, 2 \dots N$

$$\begin{aligned}
\text{Here, } 1 &= P(\Omega) \\
&= P\left(\bigcup_{i=1}^{N} \{X = x_i\}\right) \\
&= \sum_{i=1}^{N} P\left(X = x_i\right) \\
&= cN \\
\Rightarrow c &= \frac{1}{N} \quad \text{(uniform = No bias)}
\end{aligned}$$

$$\text{PMF of X is } f_x(x) = \begin{cases} \frac{1}{N} & \text{if } X = x_i, i = 1, 2 \dots N \\ 0 & \text{otherwise} \end{cases}$$

2 Moments

$$u_r' = E\left(X^r\right) = \frac{1}{N} \sum_{i=1}^{N} x_i^r$$

$$E(X) = \frac{1}{N} \sum_{i=1}^{N} x_i = \bar{x}(\text{ say })$$

$$V(X) = \frac{1}{N} \sum_{i=1}^{N} (x_i - \bar{x})^2$$

If particular if $X \sim$ Uniform $\{1, 2, \dots N\}$

$$E(N) = \frac{N(N+1)}{2}$$

$$V(N) = \frac{N^2 - 1}{12}$$

3 Problems

1. An urn contains N balls no $1, 2, \dots, N$ if n balls are randomly withdrawn in sequence each time replacing the ball selected previously. Find $P(X = k)$ and $P(Y = k), k = 1, 2, \dots N$ where X and Y is the maximum and minimum of n chosen number. Also, find $E(X)$. Show that for large $N, E(X) = \frac{nN}{n+1}$. Find $E(Y)$ as show that for large $N, E(Y) \simeq \frac{N}{n+1}$

Let X_i be the ball number drawn in the ith draw,

$$P\left(X_i = x\right) = \frac{1}{N}, \quad x = 1, 2, \ldots N$$

Clearly, X_i's are iid Uniform $\{1, 2, \ldots N\}$ as drawings are made with replacement.

$$X = \max\ X_i$$

$$\begin{aligned} F_X(x) = P(X \leqslant x) &= P\left(\max_{i=1(1)n} \{X_i\} \leqslant x\right) \\ &= P\left(X_1 \leqslant x, X_2 \leqslant x, \ldots, X_n \leqslant x\right) \\ &= \left(P\left(X_1 \leqslant x\right)\right)^n \quad (iid) \\ &= \left(F_X(x)\right)^n \\ &= \left(\sum_{i=1}^{x} P(X \leqslant i)\right)^n \\ &= \left(\sum_{i=1}^{x} \frac{1}{N}\right)^n \\ &= \left(\frac{x}{N}\right)^n, x = 1, 2, \ldots N \end{aligned}$$

$$\therefore F_x(x) = \begin{cases} 0 & , x < 1 \\ \left(\frac{[x]}{N}\right)^n & , 1 \leqslant x < N \\ 1 & , x \geqslant N \end{cases}$$

For the non-negative values of X,

$$\begin{aligned} E(X) &= \sum_{x=0}^{\infty} \{1 - F_X(x)\} \\ &= \sum_{x=0}^{N-1} \left\{1 - \left(\frac{x}{N}\right)^n\right\} + \sum_{x=N}^{\infty} (1 - 1) \\ &= N - \sum_{x=0}^{N-1} \left(\frac{x}{N}\right)^n \end{aligned}$$

For large N, $\quad \frac{1}{N} \sum_{x=0}^{N-1} \left(\frac{x}{N}\right)^n \simeq \int_0^1 u^n du = \frac{1}{n+1}$

$$\Rightarrow \sum_{x=0}^{N-1} \left(\frac{x}{N}\right)^n \simeq \frac{N}{n+1}$$

and $E(X) \simeq N - \frac{N}{n+1} = \frac{nN}{n+1}$

$$
\begin{aligned}
&\text{Here, } Y = \min_{i=1(1)n} \{X_i\} \\
&\mathrm{F}_Y(y) = P(Y \leqslant y) = 1 - P(Y > y) \\
&= 1 - P\left(\min_{i=1(1)n} \{x_i\} > y\right) \\
&= 1 - P\left(X_1 > y, X_2 > y, \ldots X_n > y\right) \\
&= 1 - \left(P\left(X_1 > y\right)\right)^n \\
&= 1 - \left(1 - F_{X_1}(y)\right)^n \\
&= 1 - (1 - \frac{y}{N})^n
\end{aligned}
$$

For the non-negative values of Y,

$$
\begin{aligned}
E(Y) &= \sum_{y=0}^{\infty} \{1 - F_Y(y)\} \\
&= \sum_{y=0}^{N-1} \left(1 - \frac{y}{N}\right)^n + \sum_{y=N-}^{\infty} (1-1) \\
&= \sum_{x=1}^{N} \left(\frac{x}{N}\right)^n, \quad x = N - y \\
&\simeq \frac{N}{n+1}, \text{ for large } N
\end{aligned}
$$

4 Repeated Trails

A trail is a single performance of a random experiment.

Suppose that we are interested in an event. A defined occurrence of A as success and non-occurrence of A as failure. When the outcome of a trail is classified as success or failure then the trail is known as Bernoulli trail.

In a sequence of trail (Bernoulli) the trails may be dependent or independent and the probability of success may remain constant or may vary from one trail to another.

A sequence of Bernoulli trail is called a repeated trails if the trails are independent and probability of success remains the constant from one trail to another.